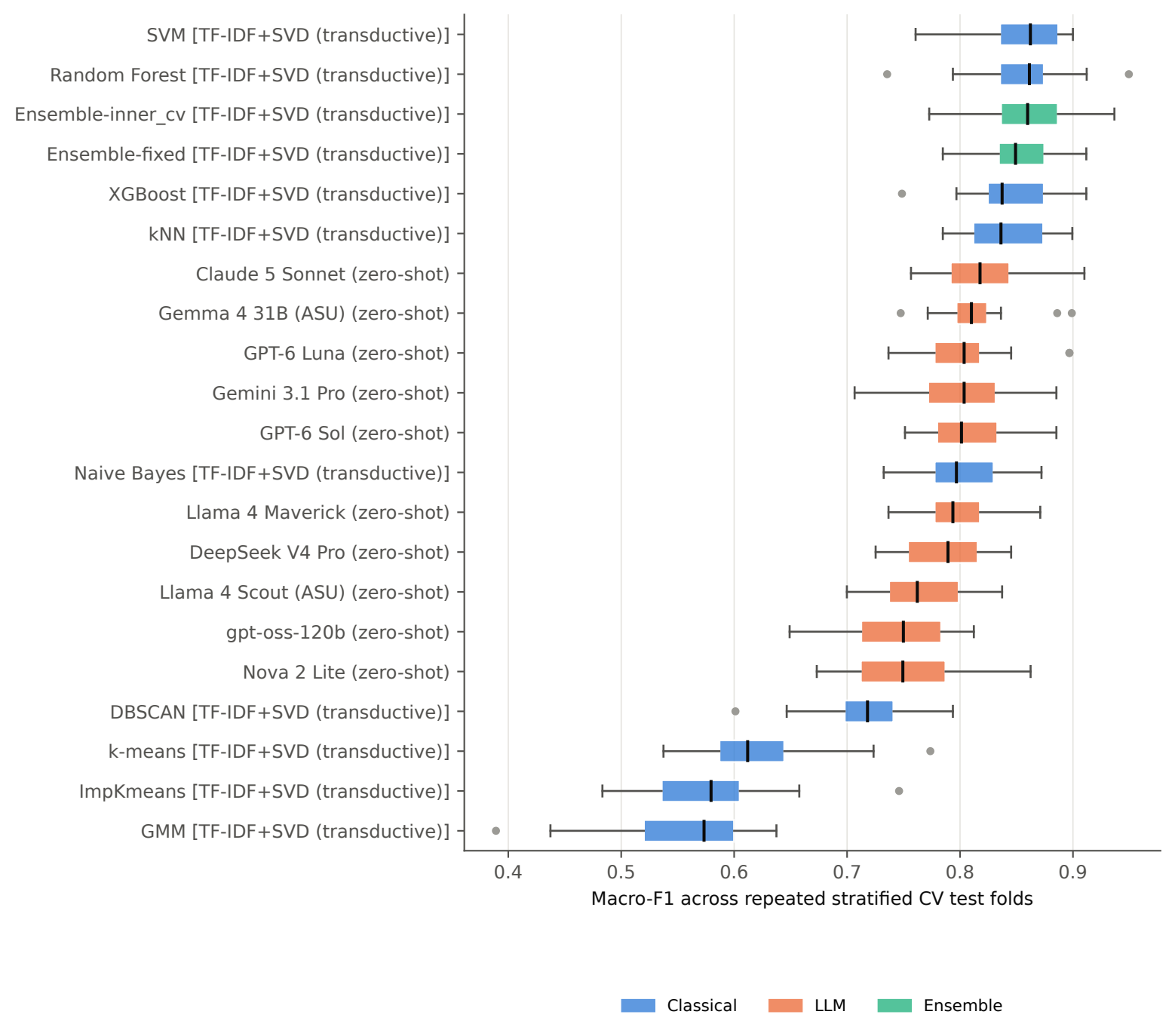
SVM [TF-IDF+SVD (transductive)]
Random Forest [TF-IDF+SVD (transductive)]
Ensemble-inner_cv [TF-IDF+SVD (transductive)]
Ensemble-fixed [TF-IDF+SVD (transductive)]
XGBoost [TF-IDF+SVD (transductive)]
kNN [TF-IDF+SVD (transductive)]
Claude 5 Sonnet (zero-shot)
Gemma 4 31B (ASU) (zero-shot)
GPT-6 Luna (zero-shot)
Gemini 3.1 Pro (zero-shot)
GPT-6 Sol (zero-shot)
Naive Bayes [TF-IDF+SVD (transductive)]
Llama 4 Maverick (zero-shot)
DeepSeek V4 Pro (zero-shot)
Llama 4 Scout (ASU) (zero-shot)
gpt-oss-120b (zero-shot)
Nova 2 Lite (zero-shot)
DBSCAN [TF-IDF+SVD (transductive)]
k-means [TF-IDF+SVD (transductive)]
ImpKmeans [TF-IDF+SVD (transductive)]
GMM [TF-IDF+SVD (transductive)]
0.4
0.5
0.6
0.7
0.8
0.9
Macro-F1 across repeated stratified CV test folds
Classical
LLM
Ensemble